Fair, Accountable, Transparent Machine Learning (FAccT ML)
ZG517

Dr. Sugata Ghosal
sugata.ghosal@pilani.bits-pilani.ac.in

BITS Pilani
Pilani Campus

innovate achieve lead

BITS Pilani
Pilani Campus

Session 1
Time – 10:40 AM to 12:40 PM

These slides are prepared by the instructor, with grateful acknowledgement of many others who made their course materials freely available online.

A little about me...

BITS Pilani
WILPD
2019-now

IBM Research –
INDIA
1998-2017

SONY Electronics, San Jose
1997-98

Amherst Systems, Buffalo, NY
1994-97

Univ. of Colorado - Denver
1993-94

Univ. of Kentucky, Lexington
MS, PhD
1988-93

Jadavpur University, Kolkata
BE
1984-88

- **I regularly teach ML, applied ML, deep learning, Advanced DL, FAccT ML @ WILPD**

innovate achieve lead

Session Content

- Objective of course
- Evaluation Plan
- Course Overview
- Bias and Fairness

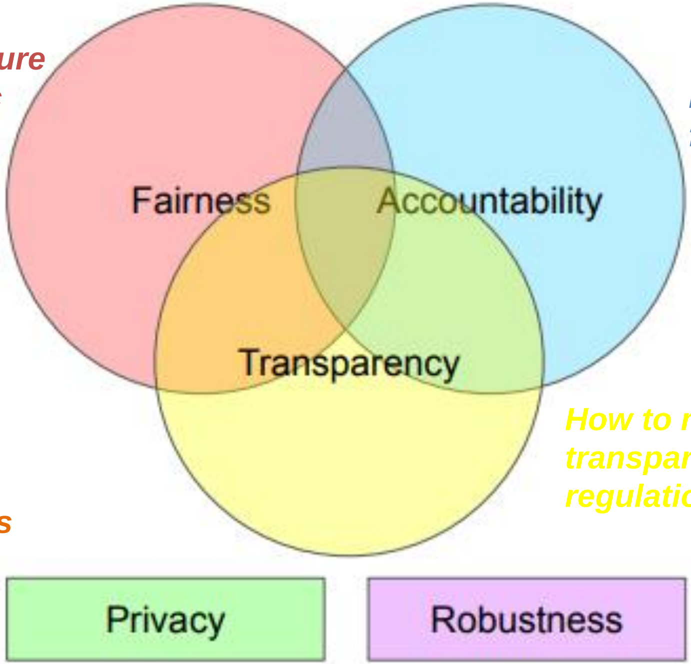
Student Domains

- Healthcare
- Language Technology
- Telecom
- Food & Beverages
- Sustainable development

What We'll Cover in this Course

What to do to ensure gender and ethnic fairness in ML models?

Who takes the responsibilities for failed ML models?



How to make ML models transparent and comply with regulations?

How to protect user privacies when exposing data to ML models?

How do we defend ML models against data poisoning?

Psychology
Social Science
Public Policy

Statistics
Theory

Machine Learning
Deep Learning

Objective of course

1. Introduce you to the concepts of bias and fairness and techniques for incorporating these in ML
2. Introduce you to the concepts of interpretability and transparency and techniques for incorporating these in ML
3. Introduce you to the concepts of robustness and techniques for robust ML
4. Introduce you to the concepts of privacy in ML

Fairness and Bias – 6 Lectures (TBD)

1. Fairness and Bias

- ✓ Sources of Bias
- ✓ Real world examples
- ✓ Sensitive Features
- ✓ Fairness through unawareness

2. Learning Fair Representations

- ✓ Major Fairness criteria
- ✓ Prejudice Removing Regularizer
- ✓ Case Studies: FICO, adult income

3. **Fairness thru input manipulation**

- ✓ Basic Data Manipulation Techniques
- ✓ Individual Fairness
- ✓ Optimized Pre-processing
- ✓ Learning to Defer

4. Fair Casual Reasoning

- ✓ Causal Fairness and Inherent Bias
- ✓ Counterfactual Fairness
- ✓ Equalized Counterfactual Odds
- ✓ Multiple Causal Worlds

5. Fairness in NLP, Computer Vision

- ✓ Biases in NLP Models
- ✓ Mitigation Strategies
- ✓ Counterfactual Face Attribution
- ✓ Gender Equalized Image Captioning
- ✓ Adversarial Removal of Gender Features

Interpretability – 5 Lectures (TBD)

- **Interpretability and Transparency**
  - ✓ ML Interpretability
  - ✓ Intrinsically Interpretable Models
  - ✓ Interpretability Concepts
  - ✓ Interpretability and performance trade-offs Instance-based Learning

- **Feature interaction for interpretability**
  - ✓ Feature Interaction
  - ✓ Layerwise Relevance Propagation
  - ✓ DeepLift
  - ✓ Shapley Additive Explanations (SHAP)

- **Example and Visualization Based Methods for Interpretability**
  - ✓ Example Based Methods
  - ✓ Counterfactual Explanations
  - ✓ Contrastive Examples
  - ✓ Concept Based Methods

- **Post Hoc Interpretability**
  - ✓ Proxy Models
  - ✓ Local Surrogate Methods, e.g., LIME
  - ✓ Rule Based Learner

- **Interpreting Deep Networks**
  - ✓ Visualization Based Methods
  - ✓ Activation Visualization
  - ✓ Gradient Based Feature Attribution

Robustness and Privacy – 3 Lectures (TBD)

- **Robustness and Adversarial Attacks & Defense**
  - ✓ Adversarial Attack
  - ✓ White-box Evasion Attack
  - ✓ Transferability of Attack
  - ✓ Black-box Evasion Attack
  - ✓ Adversarial Defense
  - ✓ Defense Strategies
  - ✓ Robust Optimization
  - ✓ Certified Defense

- **ML Auditing and Privacy**
  - ✓ ML Auditing
    - ✓ Distill-and-Compare
  - ✓ Privacy in ML
    - ✓ Differential Privacy
    - ✓ Model Inversion Attack
    - ✓ Local Differential Privacy
  - ✓ Federated Learning

Textbooks

T1	Barocas, Solon, Moritz Hardt, and Arvind Narayanan. Fairness and Machine Learning, 2018.
T2	Molnar, Christoph. Interpretable machine learning, 2019.

R1	Christopher M. Bishop, Pattern Recognition & Machine Learning, Springer
R2	Aston Zhang, Zach Lipton, Mu Li, Alex Smola, Dive into Deep Learning , 2021

Evaluation Plan (TBC)

Name	Type	Weight
2-3 Quiz, best ½ scores will be taken	Online	10%
Assignment-I	Take Home	10%
Assignment-II	Take Home	10%
Mid-Semester Test	Closed Book	30%
Comprehensive Exam	Open Book	40%

Please note there will be no change in submission dates for quiz and assignment

Lab Plan

Lab No.	Lab Objective
1	Detecting bias in ML model using different fairness metrics
2	Bias mitigation in ML model using Disparate impact remover
3	Improving the Fairness of a Classifier using Prejudice Removal Regularizer
4	Implementing LIME and SHAP algorithms to interpret different machine learning classifiers
5	Visualizing and interpreting a CNN based Image classifier
6	Implementing Adversarial attacks on CNN based Image classifier

- *Labs not graded*
- *Webinars will be conducted for lab sessions*
- *Labs will be conducted in Python*

Traditional Evaluation of ML Algorithms

- Accuracy
- Precision and recall
- Squared error
- Likelihood
- Posterior probability
- Cost / Utility
- Margin
- Entropy
- etc.

Slide credit: Pedro Domingos

Evaluating Performance

- If *y* is continuous:
  - –Sum-of-Squared-Differences (SSD) error between predicted and true *y*:

$$\mathbf{E} = \sum_{\mathbf{i}=1}^{\mathbf{n}} (\mathbf{f}(\mathbf{x}_{\mathbf{i}}) - \mathbf{y}_{\mathbf{i}})^2$$

Evaluation Metrics: Confusion Matrix

- If output y is discrete (e.g., binary) :

	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	a	b
	Class=No	c	d

a: TP (true positive) b: FN (false negative)

c: FP (false positive) d: TN (true negative)

Evaluation Metrics: Accuracy

	PREDICTED CLASS	
		Class=Yes	Class=No
ACTUAL CLASS	Class=Yes	a (TP)	b (FN)
	Class=No	c (FP)	d (TN)

- Most widely-used metric:

$$\text{Accuracy} = \frac{a+d}{a+b+c+d} = \frac{TP+TN}{TP+TN+FP+FN}$$

Class Imbalance Problem

- Lots of classification problems where the classes are skewed (more records from one class than another)
  - Credit card fraud
  - Intrusion detection
  - Defective products in manufacturing assembly line
  - COVID-19 test results on a random sample

- Key Challenge:
  - Evaluation measures such as accuracy are not well-suited for imbalanced class

Measures of Classification Performance

	PREDICTED CLASS		
ACTUAL CLASS		Yes	No
	Yes	TP	FN
	No	FP	TN

α is the probability that we reject the null hypothesis when it is true. This is a Type I error or a false positive (FP).

β is the probability that we accept the null hypothesis when it is false. This is a Type II error or a false negative (FN).

$$Accuracy = \frac{TP + TN}{TP + FN + FP + TN}$$

$$ErrorRate = 1 - accuracy$$

$$Precision = Positive\ Predictive\ Value = \frac{TP}{TP + FP}$$

$$Recall = Sensitivity = TP\ Rate = \frac{TP}{TP + FN}$$

$$Specificity = TN\ Rate = \frac{TN}{TN + FP}$$

$$FP\ Rate = \alpha = \frac{FP}{TN + FP} = 1 - specificity$$

$$FN\ Rate = \beta = \frac{FN}{FN + TP} = 1 - sensitivity$$

$$Power = sensitivity = 1 - \beta$$

ROC (Receiver Operating Characteristic)

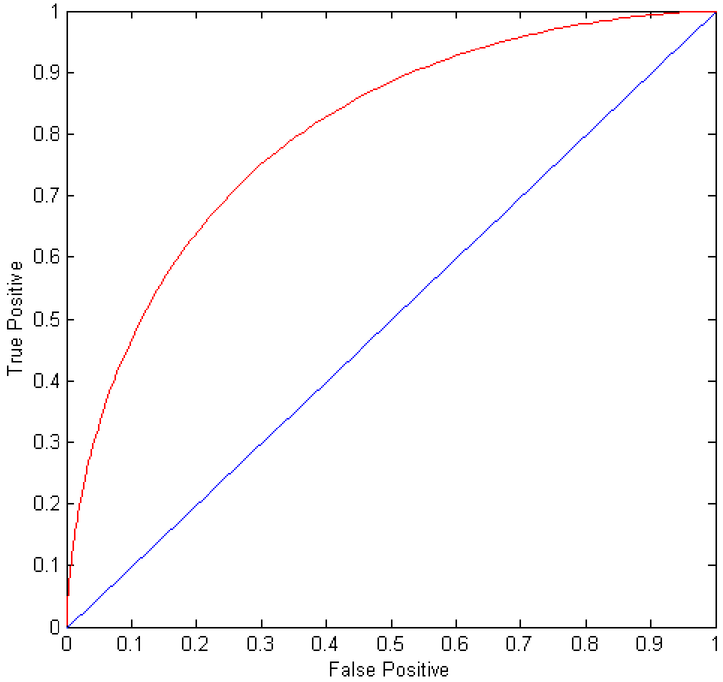
- A graphical approach for displaying trade-off between detection rate and false alarm rate
- Developed in 1950s for signal detection theory to analyze noisy signals
- ROC curve plots TPR against FPR
  - Performance of a model represented as a point in an ROC curve

ROC Curve

(TPR,FPR):

- (0,0): declare everything to be negative class
- (1,1): declare everything to be positive class
- (0,1): ideal

- Diagonal line:
  - Random guessing
  - Below diagonal line:
    - prediction is opposite of the true class



How to Construct an ROC curve

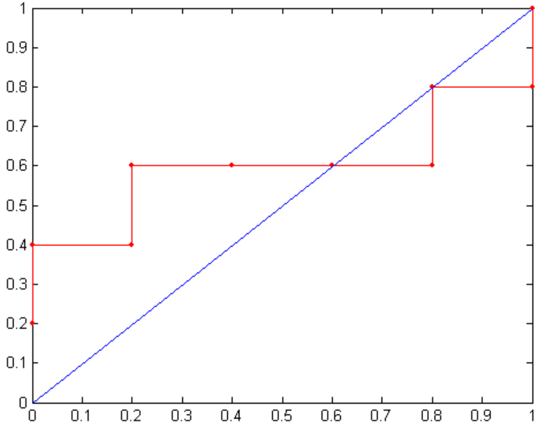
Instance	Score	True Class
1	0.95	+
2	0.93	+
3	0.87	-
4	0.85	-
5	0.85	-
6	0.85	+
7	0.76	-
8	0.53	+
9	0.43	-
10	0.25	+

- Use a classifier that produces a continuous-valued score for each instance
  - The more likely it is for the instance to be in the + class, the higher the score
- Sort the instances in decreasing order according to the score
- Apply a threshold at each unique value of the score
- Count the number of TP, FP, TN, FN at each threshold
  - TPR = TP/(TP+FN)
  - FPR = FP/(FP + TN)

How to construct an ROC curve

Class	+	-	+	-	-	-	+	-	+	+	
Threshold >=	0.25	0.43	0.53	0.76	0.85	0.85	0.85	0.87	0.93	0.95	1.00
TP	5	4	4	3	3	3	3	2	2	1	0
FP	5	5	4	4	3	2	1	1	0	0	0
TN	0	0	1	1	2	3	4	4	5	5	5
FN	0	1	1	2	2	2	2	3	3	4	5
→ TPR	1	0.8	0.8	0.6	0.6	0.6	0.6	0.4	0.4	0.2	0
→ FPR	1	1	0.8	0.8	0.6	0.4	0.2	0.2	0	0	0

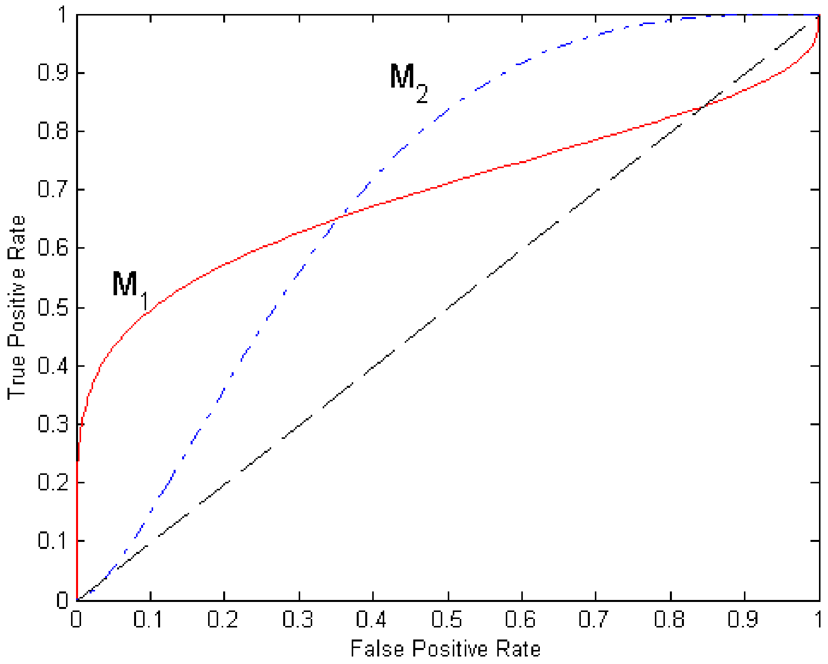
ROC Curve:



Instance	Score	True Class
1	0.95	+
2	0.93	+
3	0.87	-
4	0.85	-
5	0.85	-
6	0.85	+
7	0.76	-
8	0.53	+
9	0.43	-
10	0.25	+

- **TPR = TP/(TP+FN)**
- **FPR = FP/(FP + TN)**

Using ROC for Model Comparison



- No model consistently outperforms the other
  - M_1 is better for small FPR
  - M_2 is better for large FPR
- Area Under the ROC curve (AUC)
  - Ideal:
    - Area = 1
  - Random guess:
    - Area = 0.5

Bias and Fairness

- Fairness
  - Sources of Biases
  - Real World Examples
  - Sensitive Features

- Major Fairness Criteria
  - Fairness Through Unawareness
  - Group Fairness Metrics

ML Fairness

- What is Fairness?
  - The absence of bias towards an individual or a group (Mehrabi et al, 2019)
- Can ML Models Discriminate?
  - Aren't machines just follow human's instructions?
  - ML models approximate patterns in the data
  - Learns/Amplifies biases at the same time

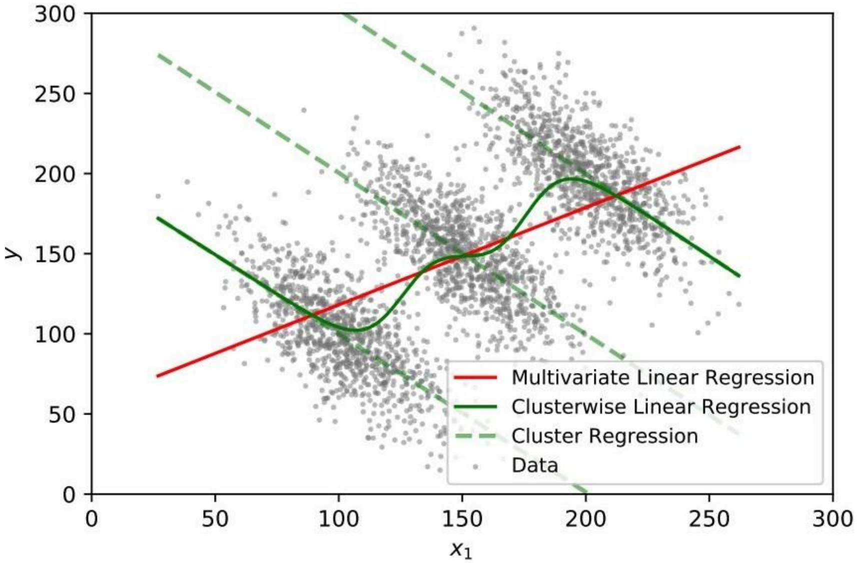
Sources of Bias

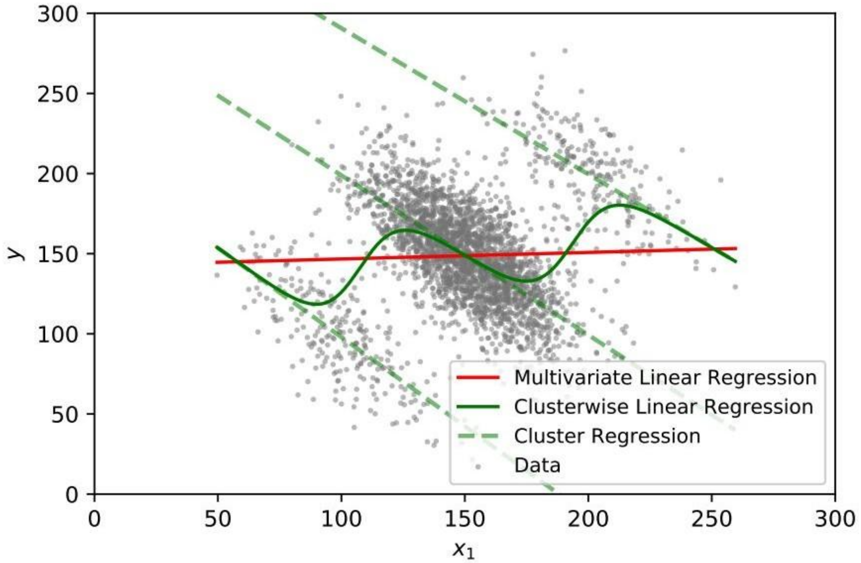
- **Data adequacy.** Infrequent and specific patterns may be down-weighted by the model and so minority records can be unfairly neglected.
  - data collection methodology can exclude or disadvantage certain groups
  - Sometimes records are removed if they contain missing values and these may be more prevalent in some groups than others.

- **Data bias or Negative Legacy.** Even if the amount of data is sufficient to represent each group, training data may reflect existing prejudices / historical unfairness
  - e.g., that female workers are paid less
  - Non-uniform data collection impacting different groups differently
  - choice of input attributes to the model

- **Model adequacy.** The model architecture may describe some groups better than others.
  - a linear model may be suitable for one group but not for another.

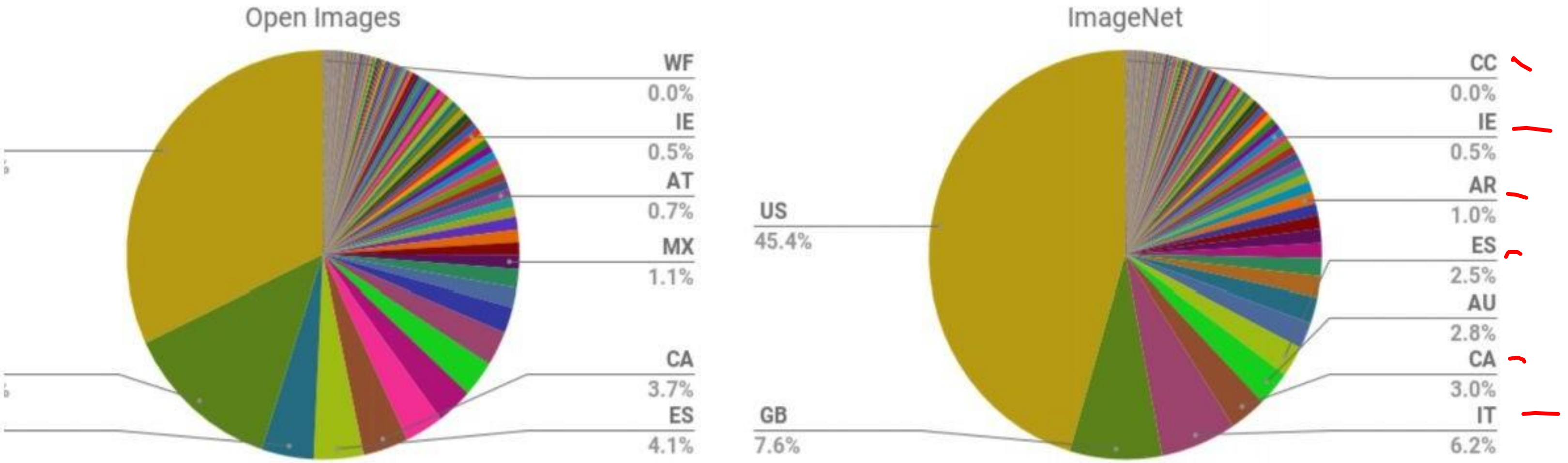
Sources of Bias





red - biased regression
dashed green - regression for each subgroup
solid green - unbiased regression

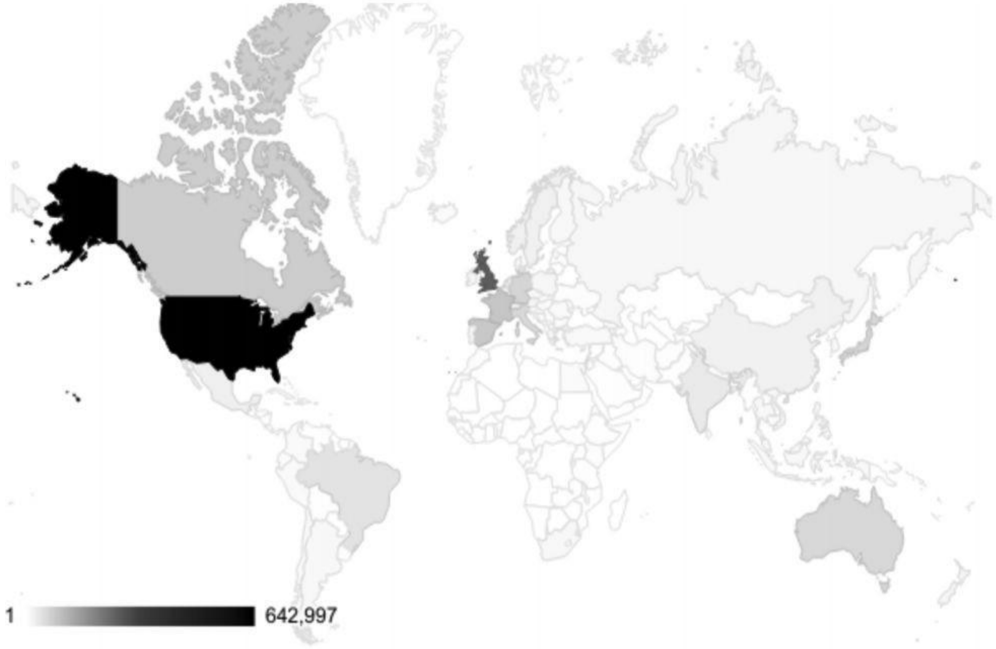
Geographical Representation of ImageNet and Open Images



[Mehrabi et al, 2019]

Geographical Representation of Open Images

- One third of the data was collected in US
- 60% of the data was from the six most represented countries.



Mehrabi et al, 2019

Graduate School Admissions to UC Berkeley, 1973

	Men		Women	
	Applicants	Admitted	Applicants	Admitted
Total	8442	**44%**	4321	35%

https://en.wikipedia.org/wiki/Simpson%27s_paradox

Graduate School Admissions to UC Berkeley, 1973

Department	Men		Women	
	Applicants	Admitted	Applicants	Admitted
A	***825***	62%	108	**82%**
B	***560***	63%	25	**68%**
C	325	**37%**	***593***	34%
D	417	33%	375	**35%**
E	191	**28%**	***393***	24%
F	373	6%	341	**7%**

https://en.wikipedia.org/wiki/Simpson%27s_paradox

Real world example of fairness

New Zealand passport robot thinks this Asian man's eyes are closed

By James Griffiths, CNN
Updated 1:46 AM ET, Fri December 9, 2016

× The photo you want to upload does not meet our criteria because:
- Subject eyes are closed

Please refer to the technical requirements.
You have 9 attempts left.

Check the photo requirements.

Read more about common photo problems and how to resolve them.

After your tenth attempt you will need to start again and re-enter the CAPTCHA security check.

Reference number: 20161206-81

Filename: Untitled.jpg

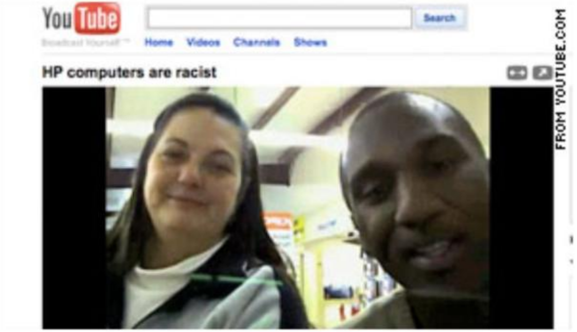
If you wish to contact us about the photo, you must provide us with the reference number given above.

New Zealand's online passport application system couldn't recognize Richard Lee's open eyes.

HP looking into claim webcams can't see black people

By **Mallory Simon**, CNN

December 23, 2009 7:25 p.m. EST



A YouTube video shows co-workers trying out an HP webcam with motion-tracking and facial recognition software.

(CNN) -- Can Hewlett-Packard's motion-tracking webcams see black people? It's a question posed on a now-viral YouTube video and the company says it's looking into it.

In the video, two co-workers take turns in front of the camera -- the webcam appears to follow Wanda Zamen as she sways in front of the screen and stays still as Desi Cryer moves about.

HP acknowledged in a statement e-mailed to CNN that the cameras may have issues with contrast recognition in certain lighting situations. The webcams, built into HP's new computers, are supposed to keep people's faces and bodies in proportion and centered on the screen as they move.

The video went viral over the weekend, garnering more than 400,000 YouTube page views and a slew of comments on Twitter.

STORY HIGHLIGHTS

- **NEW:** Video was meant to be humorous showing of software glitch, co-workers say
- Co-workers: Motion-tracking webcam moves with white woman, not black man
- "I think my blackness is

Amazon's Secret AI Hiring Tool Reportedly 'Penalized' Resumes With the Word 'Women's'

Rhett Jones
10/10/18 10:32AM • Filed to: ALGORITHMS

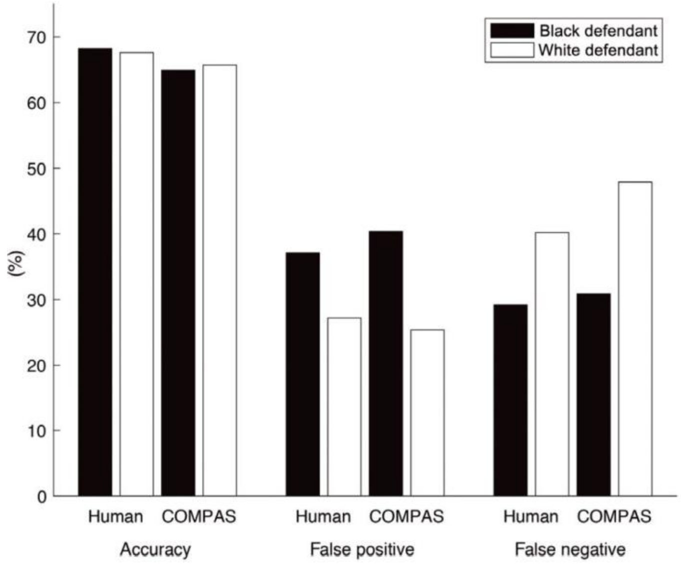
79 3

Photo: Getty

Recent Video

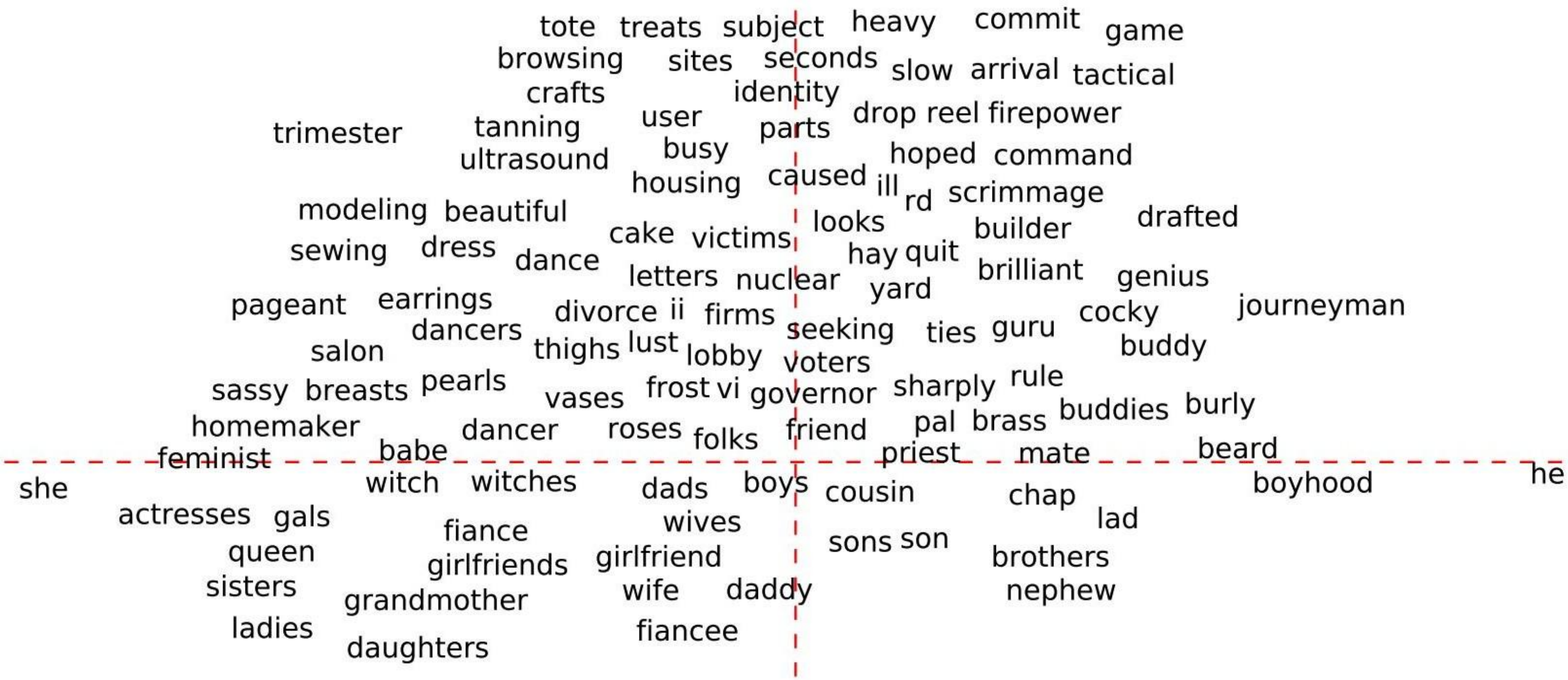
Criminal Justice (Dressel et al, 2018)

- Commercial risk assessment software known as COMPAS
  - assess more than 1 million offenders since 2000
  - predicts a defendant's risk of committing a misdemeanor or felony 137 features

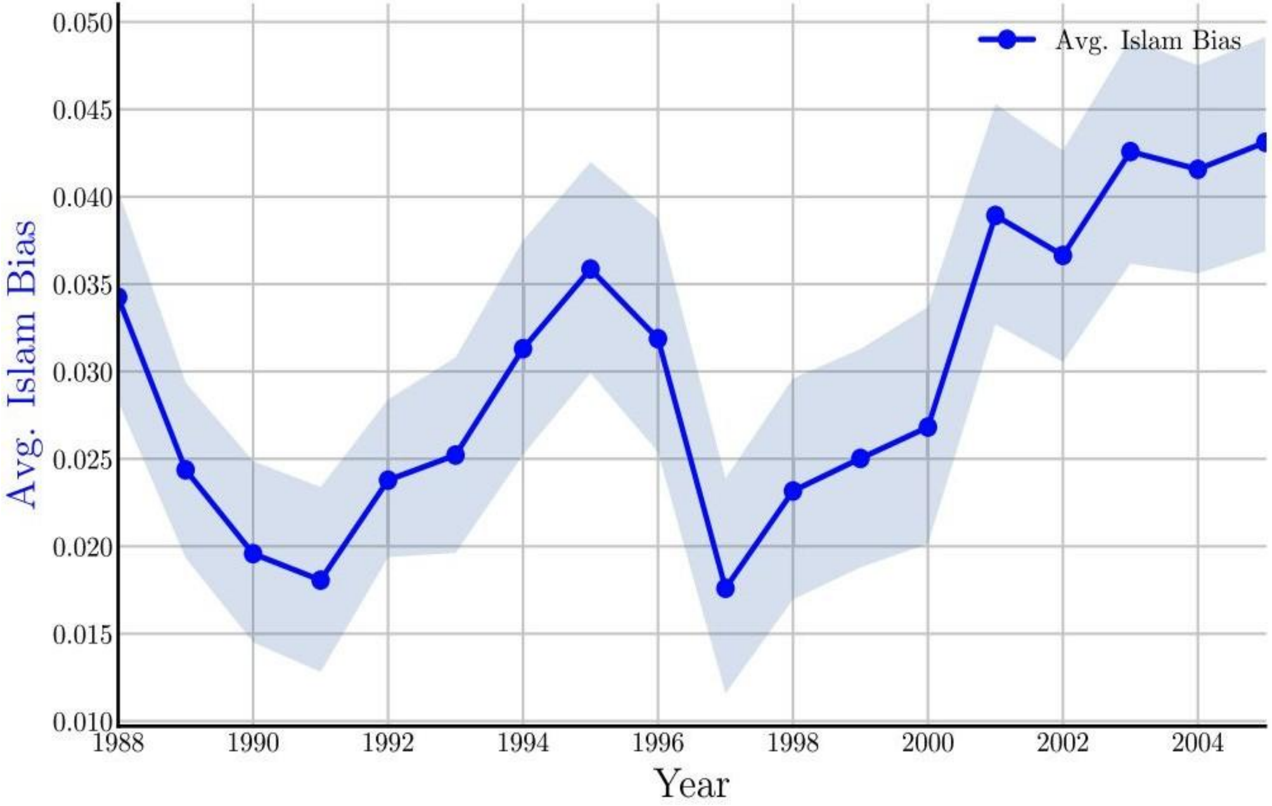


Biases in Word Embedding (Gard et al, 2018)

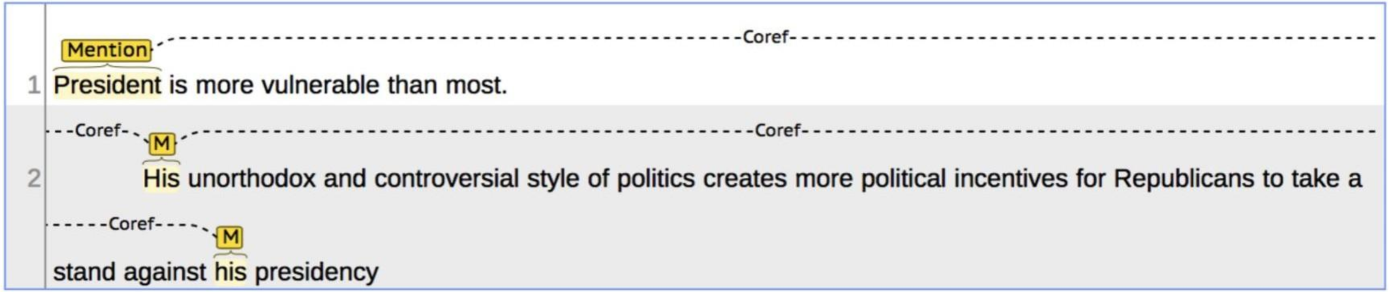
Words Projected into Gender Axes (Bolukbasi et al, 2016)



Religious Bias Related to Terrorism



Coreference Resolution (Zhao et al, 2018)



his ⇒ her

❖ Stereotypical dataset

The physician hired the secretary because he was overwhelmed with clients.

The physician hired the secretary because she was highly recommended.

❖ Anti-stereotypical dataset

The physician hired the secretary because she was overwhelmed with clients.

The physician hired the secretary because he was highly recommended.

Removing Gender Information (Wang et al, 2019)

COCO Results

imSitu Results

Sensitive (Protected) Features

- Sensitive Features
  - Identify a group
  - e.g., gender, ethnics

- Discrimination Occurs
  - When Sensitive Features Are Used Improperly
  - May lead to ML Discrimination

Protected Attributes Specified in US Fair Lending Laws

- Fair Housing Acts (FHA)
- Equal Credit Opportunity ACts (ECOA)

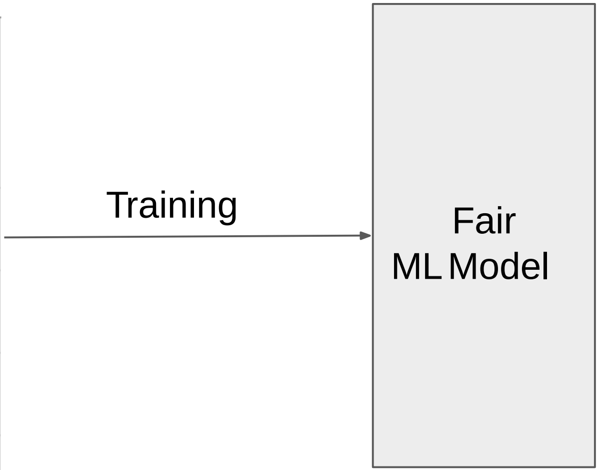
Attribute	FHA	ECOA
Race	✓	✓
Color	✓	✓
National origin	✓	✓
Religion	✓	✓
Sex	✓	✓
Familial status	✓	
Disability	✓	
Exercised rights under CCPA		✓
Marital status		✓
Recipient of public assistance		✓
Age		✓

Fairness Through Unawareness

- A ML Algorithm Achieves Fair Through Unawareness If
  - None of the sensitive features are directly used in the model

Protected

~~Race and Ethnicity~~	Skills	Years of Exp	Hired?
~~Hispanic~~	Javascript	1	no
~~Hispanic~~	C++	5	yes
~~White~~	Java	2	yes
~~White~~	C++	3	yes



Issues With Fairness Through Unawareness

Sensitive Features May Still Be Used

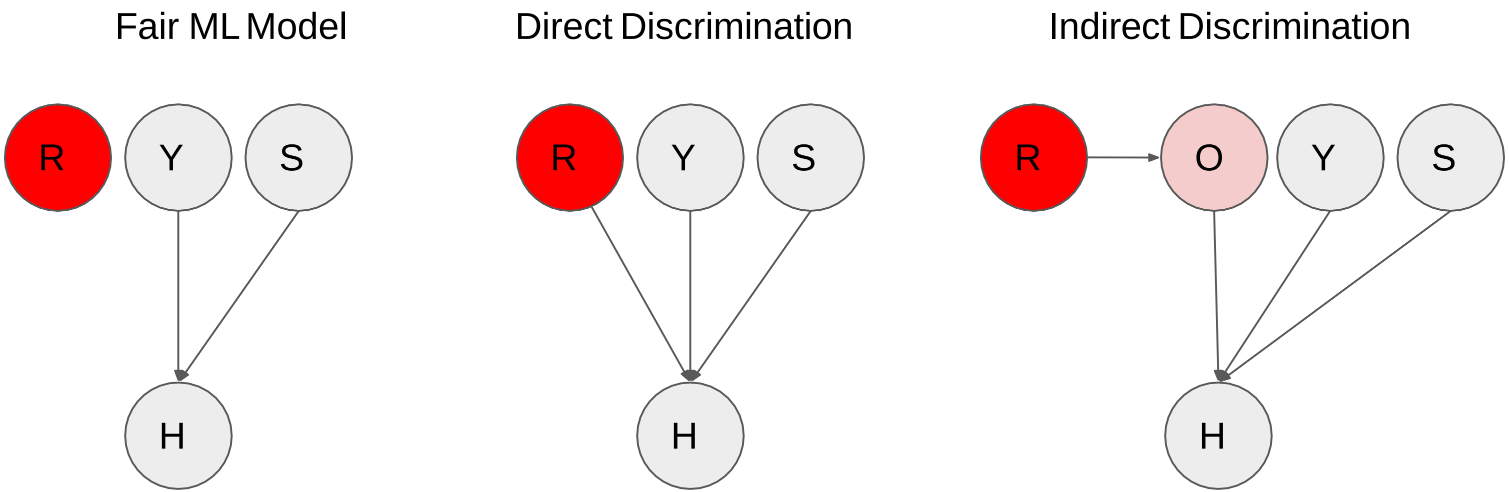
- Inferred from indirect evidence

Inferred Protected

Race and Ethnicity	Skills	Years of Exp	Often Goes to Mexican Markets	Hiring Decision
Hispanic	Javascript	1	yes	no
Hispanic	C++	5	yes	yes
White	Java	2	no	yes
White	C++	3	no	yes

Training → Discriminatory ML Model

Types of Discriminations



R - Race
Y - Years of Exp
S = Skills
O = Often Goes to Mexico Market

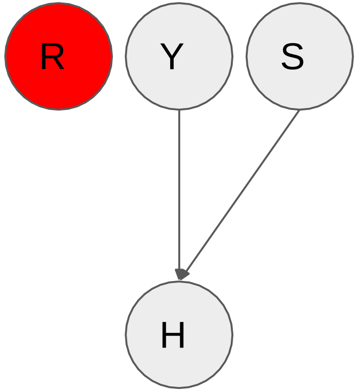
Conditions for Direct Discrimination

- A - set of protected features
- X - set of features other than protected features

- A predictor $\hat{Y}$ is direct discrimination if
  - $P(\hat{Y} \mid X, A) \neq P(\hat{Y} \mid X)$
  - i.e., $\hat{Y} \not\perp\!\!\!\perp A \mid X$

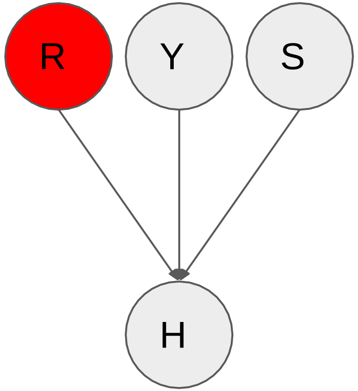
Types of Discriminations

Fair ML Model



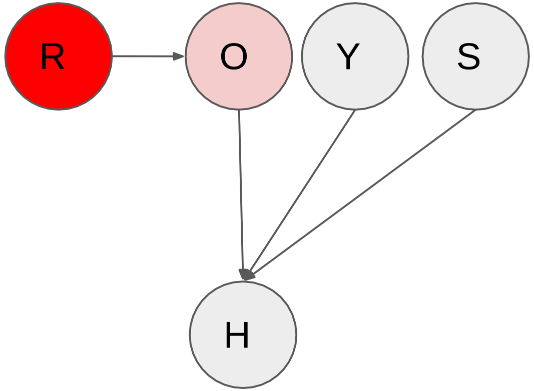
$H \perp\!\!\!\perp R \mid \{Y,S\}$
not connected

Direct Discrimination



$H \not\perp\!\!\!\perp R \mid \{Y, S\}$
(head to head)

Indirect Discrimination



$H \perp\!\!\!\perp R \mid \{Y,S, O\}$
(head to tail)

Required Reading

- Barocas: Ch 2
- Bishop: Ch 8.2
- https://www.borealisai.com/research-blogs/tutorial1-bias-and-fairness-ai/

Recommended

- Papers mentioned on the slides

Bishop, Christopher M. Pattern recognition and machine learning. springer, 2006.

Barocas, S., Hardt, M., & Narayanan, A. Fairness and Machine Learning, 2018.